Published on 11 May 2010
Licence and permissions

Effects of London dispersion on the isomerization reactions of large organic molecules: a density functional benchmark study†

Robert Huenerbein, Birgitta Schirmer, Jonas Moellmann and Stefan Grimme*

Received 5th March 2010, Accepted 10th March 2010
First published as an Advance Article on the web 11th May 2010
DOI: 10.1039/c003951a

A benchmark set of 24 isomerization reactions of large organic molecules (consisting of 24 to 81 atoms) is presented (termed ISOL). The molecules are much larger than what is typically considered in thermochemical tests. To obtain reference isomerization energies, complete basis set (CBS) extrapolations at the (SCS)-MP2 level have been computed that are augmented by perturbative third-order corrections (SCS-MP3 and MP2.5 methods). Based on these carefully examined reference data, a diverse set of common density functionals varying from GGA to double-hybrid functional level with and without dispersion correction (DFT-D) is tested. Double-hybrid and the PBE0 hybrid functionals are found to be the methods of choice for the type of main group thermochemistry examined here. For all isomerizations with an average reaction energy of 22.7 kcal mol^{-1} (in a range between 0.5 and 74.5 kcal mol^{-1}), PBE0-D, B2PLYP-D and B2GP-PLYP-D yield mean absolute deviations of 2.5, 4.1 and 2.9 kcal mol^{-1}. Most importantly it is found that the use of a dispersion correction is essential if such large molecules are considered. For all DFT methods the MAD is lowered very significantly by 1.4–5.0 kcal mol^{-1} when DFT-D is used. Intramolecular (mainly medium-range) London dispersion interactions account in some cases for more than 50% (41 kcal mol^{-1}) of the isomerization energy even though the size of the systems remains unchanged. This study also demonstrates for the first time clearly that typical DFT errors are larger than expected (about 5 kcal mol^{-1}) and that chemical accuracy (about 1 kcal mol^{-1}) even for these electronically well-behaved molecules is currently not reached by DFT. We propose this new test set as a difficult challenge for electronic structure methods that claim to be routinely applicable to large molecules. We also suggest to use a distance range resolved dispersion energy as a diagnostic for problematic cases in DFT.

1. Introduction

Applying quantum chemical methods to practical, 'real-life' problems has become a standard procedure for both, experimental and theoretical chemists. Currently, Kohn–Sham density functional theory (DFT) with its countless different functional approximations is the most widely-used *ab initio* method but nevertheless also wave function based approaches are still commonly considered especially if high accuracy is needed. All these methods have been benchmarked against thermochemical tests of various complexity.[1–3]

Within the last ten years there were major improvements of wavefunction based methods as *e.g.* the development of spin component scaled Møller–Plesset perturbation theory (SCS-MP2[4] and SCS-MP3[5]) and its variants.[6,7] Moreover, also an enormous number of new density functionals appeared. Especially the introduction of double-hybrid methods[8] and a standardized form of an empirical dispersion correction[9,10] (for earlier related work in the context of DFT see ref. 11–13) lead to an accuracy that before could only be reached by higher-level wave function based methods (for other, conceptionally somewhat different methods see *e.g.* ref. 3). However, it should also be noted that at least part of the new developments in DFT are ignored by many non-theoreticians that still use the B3-LYP/6-31G(d) level of theory even though many examples have been published where this combination dramatically fails[14–20] or seemingly good results are mainly based on (unforeseeable) error compensation.[2,15,21,22]

Judging which approach to use for a specific problem is a difficult task even for experienced computational chemists. The question if the chosen method will lead to the desired accuracy is crucial and often can only be achieved by the knowledge gained from benchmark studies. Commonly used benchmark sets are often based on atomization energies (*e.g.* G3/99,[15] G3/05[23]) which has little to do with the typical problems of *e.g.* organic chemists and thus, specialized thermochemical benchmarks and compilations thereof have been introduced.[24–27]

In 2007 we proposed[28] (based on earlier work by others[29,30]) to use 34 organic isomerization reactions (dubbed ISO34 set)

Theoretische Organische Chemie, Organisch-Chemisches Institut der Universität Münster, Corrensstraße 40, D-48149 Münster, Germany. E-mail: grimmes@uni-muenster.de; Fax: (+49)-251-8336515; Tel: (+49)-251-8336512

† Electronic supplementary information (ESI) available: Cartesian coordinates of all molecules in the ISOL set. See DOI: 10.1039/c003951a

 This journal is © the Owner Societies 2010

as a meaningfull thermochemical benchmark. It consists of molecules with less than 20 (mostly less than 10) atoms which is smaller than what is typically considered in experimental work. This important limitation is now lifted in this work and we propose a new test set consisting of 24 isomerization reactions including significantly larger molecules (termed ISOL set). The reasons to use isomerization reactions still hold: they are well defined, practically relevant, the formal size of the system is constant in the reaction and often the electronic structure changes significantly but at the same time difficult higher order effects (like vibrational anharmoniticity, relativity and core–core correlation) mostly cancel out and therefore can be neglected.

After the technical details are given in the next section, the test set is presented and the motivation for the chosen reactions is explained. Then, the accuracy of the computed reference data is discussed in great detail regarding the choice of the reference electronic structure method as well as the effect of different AO basis sets. The next part deals with the performance of wave function based methods (HF, MP2, SCS-MP2) and different DFT approximations (B-LYP, B-P86(-D), PBE(-D), B97-D, TPSS(-D), B3-LYP(-D), PBE0(-D), BMK(-D), PW6B95(-D), B2-PLYP(-D), mPW2-PLYP(-D), B2GP-PLYP(-D)). An analysis of the effects of the London dispersion energy for two selected reactions completes our investigation.

2. Methods and technical details

All DFT, HF and MP2 calculations have been performed with a modified version of TURBOMOLE 5.9.[31,32] For all MP3[5] computations the RICC[33] program developed in our group was used.

The Ahlrichs' type triple- (TZV)[34] and quadruple-zeta (QZV)[35] basis sets extended with polarization functions taken from the corresponding Dunning basis sets cc-pVXZ[36] leading to TZV(2d,2p) dubbed TZV2P and QZV(3d2f, 3p2d) dubbed QZV3P were used. Additionally, SCS-MP2 computations with the Dunning cc-pVDZ and cc-pVTZ basis sets were performed.[36]

The wavefunction based methods utilized in this work are second-order perturbation theory (MP2),[37] spin-component scaled (SCS-)MP2[4] and spin-component scaled third-order perturbation theory (SCS-MP3[5]) as well as MP2.5[38] (which essentially is 'damped' MP3, *i.e.*, $E(\mathrm{MP2.5}) = E(\mathrm{MP2}) + 1/2E^{(3)}$). Only valence electrons have been correlated in these treatments.

The range of used density functionals includes GGAs (B-LYP,[39,40] B-P86,[39,41] PBE,[42] B97-D,[10]), meta-GGAs (TPSS[43]), hybrid (B3-LYP,[39,40,44,45] PBE0[46]), meta-hybrid (BMK,[47] PW6B95[48]) and double-hybrid functionals (B2-PLYP,[8] mPW2-PLYP,[49] B2GP-PLYP[50]). Additionally the DFT methods were tested using our standard dispersion correction[10,51] (as indicated by an appended '-D').

For all GGA type functionals the resolution of the identity (RI-J) approximation[52] and auxiliary basis sets from the TURBOMOLE library were used.[53] For all perturbative treatments the RI-approximation with auxiliary basis sets that were optimized for the corresponding cc-pXZ Dunning basis[54] was used.

The structures of all molecules were optimized at the B97-D/TZVP level. Because the assessment of all methods is based on these same structures we expect no significant effect on the results by this choice.

For the following functionals we employ hitherto unpublished s_6 scaling factors[9] in DFT-D: 0.6 for PBE0 and 0.5 for PW6B95. They have been determined by fitting to the binding energies of non-covalently bound complexes (S22 set[55]).

3. Results and discussion

3.1 The ISOL set

The major aim of this work is to test quantum chemical methods on systems of similar size and complexity as "real life" problems experimental chemists use quantum chemistry for. After intensive literature searches we arrived at a set containing 24 reactions of molecules that are of typical size for organic chemistry and can be found in chemistry textbooks[56–62] or recent research papers.[63–70] It is shown in Fig. 1 and dubbed in the following and for future reference ISOL.

The molecules considered are typical and cover a wide range of structures, *e.g.* a sugar (3), a steroid (4, 7), or an organic dye (11). Only few molecules are pure hydrocarbons (9, 17). Most of them contain a few heteroatoms (*e.g.* 4, 5, 7, 9, 15), some even many of them (*e.g.* 16, 19, 22). In contrast to the small test set treated in earlier work[28] charged systems are also considered (23 and 24). Note, that in the experimental works always multiple reaction steps are necessary to achieve isomerization while we consider here for convenience only the 'end-points' of each sequence. The cartesian coordinates for all systems can be found in the ESI† and on our quantum chemistry benchmark web-site.[71]

Fig. 2 shows the root-mean-square deviation (RMSD) from our reference data (see below) averaged over all approaches under investigation (compare also Table 3) for each of the 24 reactions. These values allow a classification of the reactions as being easier or more difficult. The data show that the complexity of the reactions differs extremely. Some (like number 12 and 20) are described very accurately by nearly all methods (RMSD <1 kcal mol^{-1}). We consider reactions with an RMSD lower than 5 kcal mol^{-1} as being easy which holds for more than half of the cases. There are also very problematic reactions. The most difficult one seems to be reaction number four (RMSD $>$ 30 kcal mol^{-1}). This can be understood by looking at the corresponding structures. The reactant is a compact three-dimensional molecule while the product is an open chain. As will be discussed below in more detail, all methods that neglect or incorrectly account for intramolecular London dispersion effects will fail to describe such a reaction accurately. A similar behavior is seen for reaction number two.

3.2 Computation of reference data

No experimental heats of formation are availabe that would allow to derive reference data for this test set and the generation of theoretical isomerization energies was thus one major part of our work. The accuracy of these data is one of the most important points for a benchmark study as a doubtful

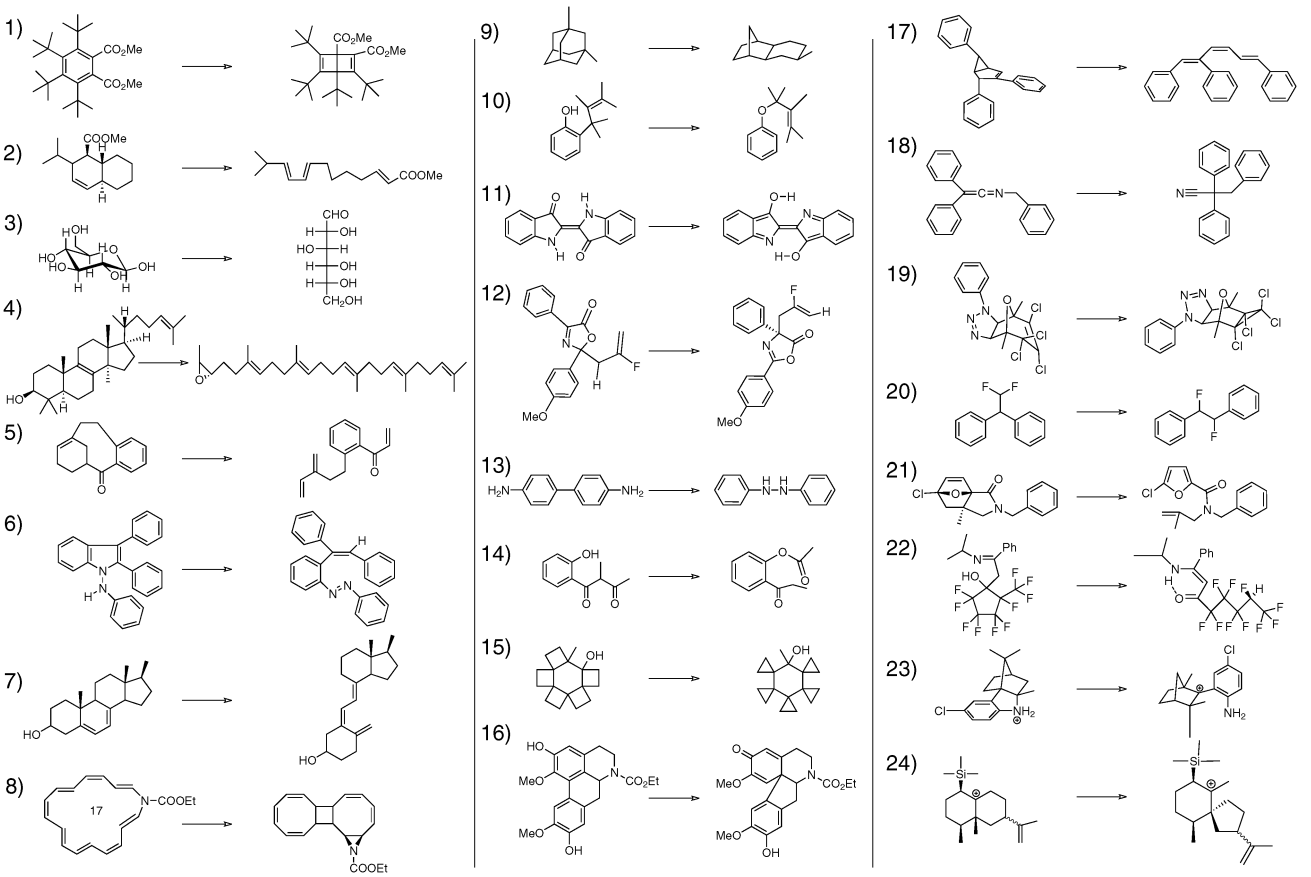


Fig. 1 The benchmark set of 24 isomerization reactions of large organic molecules.

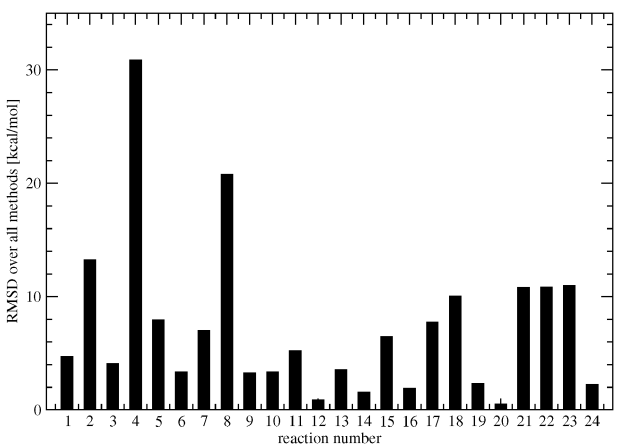


Fig. 2 RMS deviations in kcal mol^{-1} for all reactions and all tested methods (see Table 3).

reference would lead to meaningless conclusions. The best results for the ISO34 set were achieved using the current "gold standard" of quantum chemistry, CCSD(T). Unfortunately, it cannot be used for almost all molecules in this work due to the unfavorable scaling behavior of $O(N^7)$ of the computation time with system size. There is no possibility to use CCSD(T) with appropriate basis sets here and this is probably the main reason why such a benchmark has not been considered before. However, the situation is not as hopeless as it seems.

For the smaller molecules in ISO34 it has been shown that spin-component-scaled Møller–Plesset perturbation theory[4] (SCS-MP2) gives results very close to CCSD(T). Based on this experience SCS-MP2 computations with different basis sets were performed for all reactions (Table 1).

To rule out the basis set incompleteness error as far as possible, complete basis set (CBS) extrapolations according to the Helgaker scheme were performed.[72] It was originally designed to be used with correlation consistent Dunning basis sets. Computations using cc-pVDZ and cc-pVTZ were performed and extrapolated to the basis set limit (fourth column in Table 1). For most reactions the difference between the double- and the triple-zeta values is small (1–2 kcal mol^{-1}) and only in three cases it is 4–5 kcal mol^{-1} (for entries 3, 6 and 15). As expected the difference between the cc-pVTZ and the extrapolated values is even smaller. For the critical cases this difference diminishes to 2–3 kcal mol^{-1}.

Typically the next step would be an extrapolation using cc-pVTZ and cc-pVQZ as basis sets as a check. This, however, is currently impossible for this test set due to the size of some of the molecules. Nonetheless calculations with larger basis sets are needed to prove the accuracy of the reference data. Therefore another extrapolation using Ahlrichs' triple-zeta (TZV2P) and quadruple-zeta (QZV3P) sets was carried out. The QZVP set is slightly smaller than cc-pVQZ but should provide results of similar (or even better) accuracy. As can be seen in Table 1, the largest TZV2P-QZV3P difference is only around 3 kcal mol^{-1}. Furthermore the largest deviation

 This journal is © the Owner Societies 2010

Table 1 Basis set dependence and complete basis set extrapolations for SCS-MP2

	Isomerization energy/kcal mol^{-1}					
Reaction	cc-pVDZ	cc-pVTZ	CBS(D → T)	TZV2P	QZV3P	CBS(TZV2P → QZV3P)
1	67.8	68.0	67.8	71.6	70.3	69.3
2	41.6	41.3	41.8	41.1	41.0	41.8
3	4.6	9.0	10.5	8.4	10.0	10.9
4	75.4	77.3	79.8	76.4	76.0	77.4
5	36.6	34.9	34.7	35.2	35.3	35.9
6	20.4	25.3	27.1	21.4	24.6	26.2
7	19.0	19.3	19.8	18.8	18.9	19.3
8	21.6	21.7	21.2	19.8	19.6	18.8
9	23.0	23.0	23.2	22.2	22.5	22.8
10	6.9	8.0	8.5	8.0	8.2	8.5
11	37.3	36.7	36.4	36.9	37.2	36.8
12	1.0	1.1	1.1	0.7	0.8	0.9
13	30.8	35.3	36.8	35.7	35.7	35.9
14	3.2	4.6	5.4	4.0	4.6	5.3
15	8.1	0.5	−1.8	3.1	2.2	3.1
16	21.0	23.3	24.2	20.1	21.9	23.0
17	11.1	11.8	12.3	11.6	11.4	11.4
18	30.1	28.3	27.9	30.4	28.2	27.1
19	17.7	16.0	15.3	18.0	17.0	16.5
20	6.4	5.2	4.6	5.3	5.1	4.8
21	11.2	13.0	14.0	13.9	13.0	13.3
22	−1.9	−1.1	−1.3	−0.3	−0.3	0.0
23	27.6	28.0	28.4	27.8	27.8	28.2
24	16.4	16.1	16.0	15.9	16.0	16.1

between the quadruple-zeta and the extrapolated results (T → Q) is only about 1.6 kcal mol^{-1} (reaction 6) and most deviations are significantly smaller than 1 kcal mol^{-1}. There is also good agreement between D-T(Dunning) and T-Q(Ahlrichs) extrapolations. A conservative error estimate with respect to the SCS-MP2 basis set limit of the T-Q(Ahlrichs) data (that will be used exclusively in the following) is about 1–2 kcal mol^{-1}.

Together with the basis set the applied quantum chemical method plays the most important role for generating highly accurate reference data. Even though SCS-MP2 performed excellent for the ISO34 set the applicability for the larger molecules examined here has to be proven. In the standard procedure the basis set limit is extrapolated *via* SCS-MP2 or MP2 and then a higher-level correction (normally CCSD(T)) using a small basis set is added.[55] This approach is based on the fact that higher-order correlation effects usually have only a weak basis set dependence. We here employ the SCS-MP2 variant

$$E(\text{high-level})/\text{CBS} = E(\text{SCS-MP2})/\text{CBS} + [(E(\text{high-level}) - E(\text{SCS-MP2})]/\text{small-basis}. \quad (1)$$

The problem with this procedure is that as mentioned above CCSD(T) as the high-level results cannot be obtained within a reasonable amount of time. Thus SCS-MP3 was chosen as alternative in which the last term on the right side of the above equation is given by the third-order energy correction $E^{(3)}$. Analogously, substitution of E(SCS-MP2) by E(MP2) and the higher-order correction by $\frac{1}{2}E^{(3)}$ yields MP2.5.[38] The results, which here are based on the SV(P) valence double-zeta expansion as the small basis can be found in Table 2, a graphical comparison is given in Fig. 3.

In general the third-order correction to the reaction energy is small (2–3 kcal mol^{-1} or less) and for most reactions, both schemes (SCS-MP2 and MP2.5) deviate by less than 1–2 kcal mol^{-1} from each other. Unfortunately, there is one extreme outlier: for reaction four both methods differ by about 10 kcal mol^{-1} which we consider as a too large uncertainty for a reference value. By mischance this special reaction consists of the largest molecules of the entire test set which makes a CCSD(T) computation impossible. The other outlier is reaction two with a discrepancy of about 5 kcal mol^{-1}. This reaction is very similar to reaction four as can be seen from Fig. 1: the reactants are ring systems while the products are

Table 2 Computed isomerization energies at SCS-MP2/CBS and estimated SCS-MP3/CBS and MP2.5/CBS levels

Reaction	SCS-MP2/CBS	SCS-MP3/CBS	MP2.5/CBS
	Isomerization energy/kcal mol^{-1}		
1	69.3	68.7	71.6
2	41.8	40.6	45.4
3	10.9	11.7	13.3
4	77.4	74.5	85.9
5	35.9	34.9	38.1
6	26.2	25.9	29.9
7	19.3	18.8	21.6
8	18.8	18.3	17.2
9	22.8	22.3	23.0
10	8.5	7.9	9.3
11	36.8	38.1	36.0
12	0.9	1.0	0.9
13	35.9	35.1	35.9
14	5.3	5.2	6.5
15	3.1	3.9	1.1
16	23.0	22.6	27.7
17	11.4	11.1	13.3
18	27.2	26.1	24.7
19	16.5	17.0	16.2
20	4.8	4.5	4.7
21	13.3	13.4	14.1
22	0.0	0.5	2.8
23	28.2	27.6	29.8
24	16.1	16.1	16.5

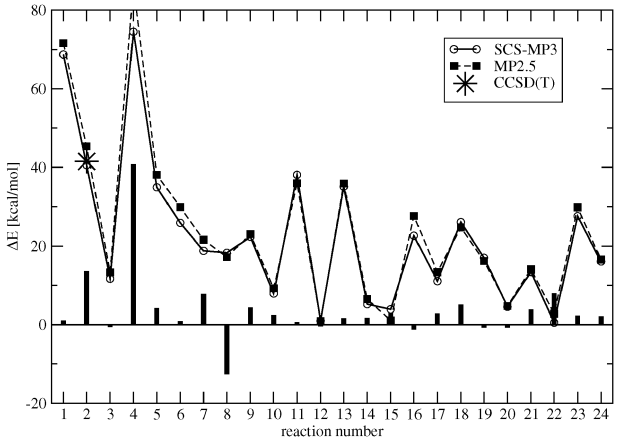


Fig. 3 Isomerization reaction energies for SCS-MP3/CBS and MP2.5/CBS. The bars show the unscaled dispersion correction for comparison. The lines are just drawn to guide the eye.

chains, the main difference is the system size. Note that there is a major influence of London dispersion in both reactions (see Fig. 3). For reaction two, CCSD(T)/SV(P) computations are feasible which allows to substitute the $\frac{1}{4}E^{(3)}$ term by the CCSD(T)-MP2 correction. As can be seen in Fig. 3 the agreement between the estimated SCS-MP3/CBS and the estimated CCSD(T)/CBS results is excellent (about 1 kcal mol^{-1}). The deviation to the MP2.5 result is larger. The similarity of reactions two and four leads to the conclusion that also for reaction four SCS-MP3/CBS is more trustworthy than MP2.5.

The above considerations and previous experience[5,28] lead us to the decision to use estimated SCS-MP3/CBS values instead of SCS-MP2/CBS or MP2.5/CBS as reference. SCS-MP3 also yields a very small MAD for the ISO34 set (0.9 kcal mol^{-1}). Including the basis set extrapolation error we estimate the maximum total error of SCS-MP3/CBS for this test set to be less than ±4–5 kcal mol^{-1}. For most reactions it should be significantly smaller. As will be shown below, most DFT methods perform significantly worse so that this uncertainty is small enough for an assessment of these more approximate methods. However, it is also noted that the best (dispersion corrected) density functionals with mean absolute deviations (MAD) of 2–3 kcal mol^{-1} come close to the accuracy of the reference data which should be taken into account in their evaluation.

3.3 Assessment of quantum chemical methods

In order to minimise the influence of the basis set, the large QZV3P basis was used in the following. This comprises from 984 to 3017 contracted AO basis functions for the systems in our set. These values diminish to 420–1256 functions when the less accurate but still recommended TZV2P set (*e.g.* for GGAs or hybrid-GGAs) is used. The most important statistical data together with a classification of the methods are given in Table 3. In contrast to ISO34, where deviations of more than 3 kcal mol^{-1} were considered as outliers, here 10 kcal mol^{-1} was chosen as a limit owing to the fact that the absolute values of most isomerization energies are larger and the reference values are slightly more uncertain than for the small molecules. The entries are ordered by accuracy (according to the resulting RMSD). Whenever the empirical dispersion correction was used a "-D" is suffixed to the functional name.

Based on previous experience it is not surprising that SCS-MP2 clearly outperforms all other methods. One should, however, keep in mind that the reference is based on SCS-MP2 values and that the higher-order corrections are small. The maximum deviation (MAXD) of 2 kcal mol^{-1} is within the estimated error of the reference data. The situation is different for MP2. The largest deviation is 16 kcal mol^{-1} for reaction four, which is considered as being very large for an absolute reaction energy of 69 kcal mol^{-1}. This can be explained by the well known tendency of MP2 to overestimate dispersion.[73,74] The Hartree–Fock (HF) results are only printed for reasons of completeness. An RMSD of more than 14 kcal mol^{-1} and a MAXD of more than 50 kcal mol^{-1} disqualifies HF for any practical purpose and also shows how important electron correlation is for our problem.

The following discussion is divided into two parts. In the first standard DFT functionals without dispersion correction are discussed while the second part deals with the influence of the D-correction. The performance of standard (uncorrected) DFT functionals varies from disastrous to moderate at best. B-LYP shows an RMSD of more than 20 kcal mol^{-1}, has more than 10 outliers and a MAXD of more than 70 kcal mol^{-1}. Note that this is even worse compared to uncorrelated Hartree–Fock. The popular hybrid functional B3-LYP performs slightly better but st ill the errors are enormous (RMSD = 16.8 kcal mol^{-1} MAXD = 57 kcal mol^{-1}). Significant improvement can be seen when going to double-hybrid functionals B2-PLYP and B2GP-PLYP. Especially Martin's B2GP-PLYP[50] outperforms all other non-dispersion-corrected approaches (RMSD = 3.9 kcal mol^{-1}, MAXD = 11.5 kcal mol^{-1}) but even with this functional reaction four is an outlier. An interesting finding is that even the GGA functionals B-P86 and PBE top the often used (and computationally more demanding) B3-LYP. The hybrid PBE0 even halves the errors of B3-LYP. Comparing PBE to its meta-GGA counterpart TPSS we observe no improvement which is similar to the conclusions for the ISO34 set. Contrary, the meta-hybrid functional PW6B95 yields results close to the double-hybrid ones.

The relative order of the performance for the dispersion corrected functionals is similar to the one without the correction whereas the absolute deviations are of different magnitude. The most surprising outcome is the excellent performance of PBE0-D. This dispersion corrected hybrid functional outperforms all other methods with an MAD of 2.5 kcal mol^{-1} which is on the order of the estimated accuracy of the reference. The accuracy of the double-hybrid functional B2GP-PLYP-D is similar (MAD = 2.9 kcal mol^{-1}). The other two dispersion corrected double-hybrids B2-PLYP-D and mPW2-PLYP-D are slightly worse (MAD of about 4.1 kcal mol^{-1}) but still only one or two outliers are found. Comparing B2-PLYP and mPW2-PLYP it is observed that mPW2-PLYP performs better when no dispersion correction is used but B2-PLYP gets ahead as soon as the correction is applied.[51] Similar high accuracy is achieved using the meta-hybrid PW6B95-D (MAD = 3.1 kcal mol^{-1}). The tremendous

 This journal is © the Owner Societies 2010

Table 3 Statistical assessment of quantum chemical methods using the QZV3P basis set (in kcal mol^{-1}). The methods are ordered according to decreasing accuracy

	Type[a]	RMSD[b]	MAD[c]	MAXD[d]	Outliers[e]
SCS-MP2	WFT	0.95	0.74	2.0 (18)	0
PBE0-D	hybrid GGA (25%)	3.39	2.53	9.1 (18)	0
B2GP-PLYP-D	double-hybrid GGA (65%)	3.92	2.88	11.5 (8)	1
MP2	WFT	4.39	3.00	15.9 (4)	1
BMK-D	hybrid meta-GGA (42%)	4.67	3.18	17.1 (15)	1
PW6B95-D	hybrid meta-GGA (28%)	5.03	3.09	11.9 (8)	1
B2-PLYP-D	double-hybrid GGA (53%)	5.49	4.08	15.5 (8)	1
mPW2-PLYP-D	double-hybrid GGA (55%)	5.62	4.14	15.3 (8)	2
BMK	hybrid meta-GGA (42%)	6.34	4.61	18.4 (4)	2
B-P86-D	GGA	6.55	4.65	18.5 (8)	4
PBE-D	GGA	6.65	4.71	17.7 (8)	4
B2GP-PLYP	double-hybrid GGA (65%)	6.78	4.71	20.1 (4)	2
TPSS-D	meta-GGA	7.08	5.49	19.2 (8)	4
PBE0	hybrid GGA (25%)	7.41	4.93	25.1 (4)	3
B3-LYP-D	hybrid GGA (20%)	8.10	6.03	20.3 (8)	5
mPW2-PLYP	double-hybrid GGA (55%)	8.76	6.07	26.9 (4)	4
PW6B95	hybrid meta-GGA (28%)	9.05	6.34	29.0 (4)	5
B2-PLYP	double-hybrid GGA (53%)	9.72	6.71	30.3 (4)	5
B97-D	GGA	11.46	8.61	26.0 (8)	9
PBE	GGA	12.15	7.97	39.4 (4)	7
HF	WFT	14.30	9.14	55.9 (4)	8
B-P86	GGA	14.43	9.59	47.1 (4)	8
TPSS	meta-GGA	15.18	10.19	51.0 (4)	9
B3-LYP	hybrid GGA (20%)	16.72	11.04	56.8 (4)	10
B-LYP	GGA	22.28	14.84	74.0 (4)	10

[a] WFT: wave function theory; GGA: generalized gradient approximation (non-hybrid) functional. The number in parentheses is the admixture of nonlocal HF exchange in a hybrid functional. Meta functionals include kinetic energy densities (occupied orbital-dependent quantities). [b] Root-mean-square deviation. [c] Mean absolute (unsigned) deviation. [d] Maximum absolute (unsigned) deviation. Corresponding reaction number in parentheses. [e] Number of absolute deviations $>$ 10 kcal mol^{-1}.

importance of dispersion becomes also obvious considering that a dispersion corrected GGA functional (PBE-D) yields results similar to the best non-corrected double-hybrid functional currently available (B2GP-PLYP). The functional BMK(-D) is a special case. The overall statistical indicators are very good (MAD = 3.2 kcal mol^{-1}) but we nevertheless would not recommend it. The problem is the MAXD of 17.1 kcal mol^{-1} for reaction number 15. As can be seen from Fig. 2 this reaction is not problematic for almost all other methods. This somewhat unpredictable behavior seems to indicate overfitting and limits the applicability of BMK in practice.

As can be concluded from the above results all DFT functionals suffer from the inability to describe London dispersion effects (although to a very different degree) which seem to be surprisingly large here although formally the sizes of the molecules remain the same in the isomerization. Reaction four which is extremely influenced by dispersion effects (by about 41 kcal mol^{-1}) is the most problematic case for all functionals. Nevertheless, the formation of a steroid can be considered as a quite normal problem of organic chemistry. We consider the finding that merely the change of molecular shape has a rather dramatic effect on medium- and long-range correlations as being very important for theoretical thermochemistry.

3.4 Difficulty of the test set

In order to compare the difficulty of the smaller ISO34 and the new ISOL test set the MADs of all density functionals with and without dispersion correction are plotted in Fig. 4.

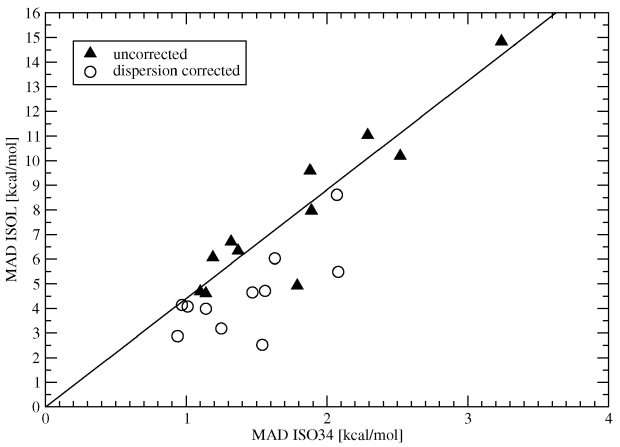


Fig. 4 Comparison of the MADs (in kcal mol^{-1}) for the ISO34 and the ISOL test set for uncorrected and dispersion corrected density functionals. Linear regression for uncorrected functionals yields a slope of 4.4 and an intercept of zero.

For the dispersion uncorrected functionals there is a correlation of the MADs which could be described reasonably well by a linear regression line with a slope of about four. This shows that the reactions of the large molecule set are by far more difficult to describe than those in ISO34. The dispersion corrected functionals do not show any systematic behavior upon enlargement of the molecules and the distribution of data points seems to be more or less randomly scattered. These observations are in agreement with the above finding that dispersion causes the major problem. Furthermore it can be

concluded that one cannot really estimate the performance of a density functional for larger molecules solely based on a benchmark of much smaller systems. This in particular holds for the lower-level (GGA) functionals while the double-hybrid methods (and also the WF based techniques not shown) behave more predictable. This is also an important new outcome of the present study.

3.5 Analysis of the dispersion contribution

For the most problematic reaction number four that is extremely influenced by the dispersion correction we performed a further analysis in order to get more insight into the topic. For comparison we also provide data for reaction 12 which is less problematic. For both molecules in the reactions, the contribution of each atom pair to the total dispersion energy is partitioned to a certain interval as defined by their interatomic distance. We have used binning intervals of 0.2–1.0 Å and distributed the dispersion energy to 15 intervals in the range <2 Å to 10 Å. The results are shown in Fig. 5.

Rather surprisingly, the dispersion energy of the isomers in reaction four is very similar for distances larger than about 5 Å meaning that the change of shape of the molecules has only a small influence in this respect. Significantly higher (absolute) dispersion energies for the more compact reactant structure are found in intervals <4 Å. The largest stabilization (-27 kcal mol^{-1}) of the reactant results in a range 2.5–3.5 Å which amounts to about two thirds of the total dispersion energy change. This finding is in agreement with the results from our previous study of alkane branching[16] which suggests that the change of the correlation energy at medium distances (1,3-interactions in the terminology of organic chemists, termed 'overlap-dispersive' in ref. 75) is a dominant factor. On the contrary, for the unproblematic reaction 12, the main differences between the two isomers occur for more typical vdW distances between 3 and 5 Å. The large changes in reaction four at much smaller distances result from the many carbon atoms with sp^2 hybridization (with mostly only two carbon neighbors) that are transformed to higher coordinated carbon atoms in the product. This changes the correlation energy considerably (as shown by local SCS-MP2 in ref. 16) and we think that this conclusion is rather general and likely holds for many organic reactions. The analysis of the range separated dispersion energy seems to be a practical tool to detect in this respect problematic cases for DFT.

4 Summary and conclusion

Often new quantum chemical methods are tested only on molecules containing very few atoms. We consider even medium-sized systems with about 20 atoms as *e.g.* included in the G3 thermochemical benchmark as rather small compared to "real" molecules that are often considered in practice. The new thermochemical benchmark presented here tries to close this gap. It is based on the (seemingly trivial but nonetheless not widely known or accepted) insight that intramolecular London dispersion effects become very significant for larger systems.

Our new test set contains 24 reactions of 48 large molecules consisting of 24 to 81 atoms. The molecules are taken without any modifications from synthetic organic chemistry. Because no experimental thermochemical data are availabe, state-of-the-art calculations were performed to obtain reasonably accurate reference data. Based on a careful analysis of the data provided by different approaches, we haven chosen SCS-MP3/CBS as a viable reference method. Its maximum estimated errors are less than about 4–5 kcal mol^{-1} and typical errors are likely to be smaller than 2 kcal mol^{-1}. This assessment is supported by basis set extrapolations using basis sets of different quality and the comparison of SCS-MP2, MP2.5 and SCS-MP3 results.

A diverse set of density functionals, *i.e.*, (meta-)GGA, (meta-)hybrid and double-hybrid functionals with and without dispersion correction was tested on these reactions. The findings allow some general conclusions:

For large systems the contribution of the intramolecular London dispersion energy is crucial. Even for double-hybrid functionals that partially account for these effects through their virtual orbital dependent terms, insufficient accuracy is obtained. Consistent with this conclusion is the also only moderate performance of MP2 which often overestimates

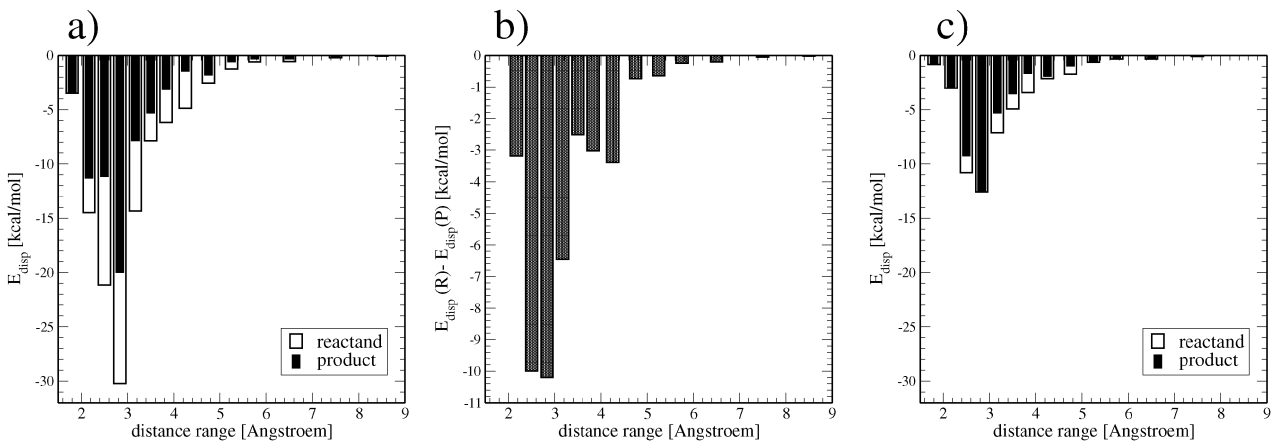


Fig. 5 Range separated dispersion energies (unscaled ($s_6 = 1$), bin widths of 0.3–1.0 Å, in kcal mol^{-1}). The bar is always placed in the middle of the considered interval. (a) Total values for the reagent and the product of reaction four. (b) corresponding difference, *i.e.*, the range separated dispersion contribution to the isomerization energy. (c) same as (a) but for the unproblematic reaction number 12.

 This journal is © the Owner Societies 2010

the relevant correlation effects. Our finding that merely the change of molecular shape and rather trivial bond rearrangements have a dramatic effect on medium-range correlations seems to be an important general conclusion for theoretical thermochemistry. These electron correlations are most problematic for many density functionals while the more long-range effects are easily handled by a semi-classical dispersion correction. If such effects should run under the header 'London dispersion' is merely a (difficult) semantic question because their character changes continuously with interatomic distance.

For all functional classes, the simple standard DFT-D correction scheme provides a large, in some cases dramatic improvement of the accuracy. We strongly recommend its general use in thermochemical applications even if no van der Waals complexes are involved and intramolecular dispersion effects are not obvious from structural considerations.

The double-hybrid functionals have again proven their general applicability compared to older standard functionals. All (dispersion corrected) variants yield excellent results. The most recent of the tested functionals, B2GP-PLYP, clearly outperforms the older variants B2-PLYP and mPW2-PLYP. This is in line with recent experience also from other groups[76] that the accuracy for main group thermochemistry improves when the contribution of non-local terms is increased.

The picture for the hybrid functionals is also in line with previous experience. PBE0-D is the best overall functional for this test set and clearly outperforms B3-LYP-D. Also as noted before, the meta-GGA approach in TPSS introduces no improvement compared to PBE. As a by-product of our study we found that the (previously overlooked) PW6B95 meta-hybrid GGA yields very good results similar to the double-hybrids.

Even though for this test set PBE0-D performs surprisingly well, our final recommendation for general applications are the double-hybrids. This type of functional has proven higher accuracy in various applications (robustness) and this is clearly needed for research on systems that are not already benchmarked. It is also better for reaction barriers that are important in general chemistry applications. For the ISOL set the deviations provided by B2GP-PLYP-D are on the order of the accuracy of the reference. The higher computational cost compared to conventional hybrid functionals is small as even for the largest systems investigated here the rate determining step is the solution of the Kohn–Sham-SCF equations and not the computation of the perturbative correction. We strongly recommend the ISOL benchmark for future evaluations of new quantum chemical methods and consider it as a challenge for methods that claim to be routinely applicable to large molecules. Also other approaches to the dispersion and mid-range correlation problem in DFT could be benchmarked on this set. It might also be useful for testing the accuracy of non-DFT methods that typically truncate in various ways a wavefunction expansion.

Acknowledgements

This work has been financially supported by the Deutsche Forschungsgemeinschaft in the framework of the SFB 858, the Forschergruppe 'Unconventional Approaches for the Activation of Hydrogen' (grant GR1927/4-1), and by the NRW Graduate School of Chemistry through a scholarship to R.H.

References

1 *Theoretical Thermochemistry*, ed. J. Cioslowski, Kluwer Academic Publishers, New York, 2001.
2 J. Cioslowski, M. Schimeczek, G. Liu and V. Stoyanov, *J. Chem. Phys.*, 2000, **113**, 9377–9389.
3 Y. Zhao and D. G. Truhlar, *Acc. Chem. Res.*, 2008, **41**, 157–167.
4 S. Grimme, *J. Chem. Phys.*, 2003, **118**, 9095–9102.
5 S. Grimme, *J. Comput. Chem.*, 2003, **24**, 1529–1537.
6 Y. Jung, R. C. Lochan, A. D. Dutoi and M. Head-Gordon, *J. Chem. Phys.*, 2004, **121**, 9793–9802.
7 T. Takatani, E. G. Hohenstein and C. D. Sherrill, *J. Chem. Phys.*, 2008, **128**, 124111.
8 S. Grimme, *J. Chem. Phys.*, 2006, **124**, 034108–034116.
9 S. Grimme, *J. Comput. Chem.*, 2004, **25**, 1463–1473.
10 S. Grimme, *J. Comput. Chem.*, 2006, **27**, 1787–1799.
11 F. A. Gianturco and F. Paesani, *J. Chem. Phys.*, 2000, **113**, 3011–3019.
12 Q. Wu and W. Yang, *J. Chem. Phys.*, 2002, **116**, 515–524.
13 M. Elstner, P. Hobza, T. Frauenheim, S. Suhai and E. Kaxiras, *J. Chem. Phys.*, 2001, **114**, 5149–5155.
14 P. C. Redfern, P. Zapol, L. A. Curtiss and K. Raghavachari, *J. Phys. Chem. A*, 2000, **104**, 5850–5854.
15 L. A. Curtiss, K. Raghavachari, P. C. Redfern and J. A. Pople, *J. Chem. Phys.*, 2000, **112**, 7374–7383.
16 S. Grimme, *Angew. Chem., Int. Ed.*, 2006, **45**, 4460–4464.
17 M. D. Wodrich, C. Corminboeuf and P. von Rague Schleyer, *Org. Lett.*, 2006, **8**, 3631–3634.
18 P. R. Schreiner, A. A. Fokin, Robert A. Pascal, Jr and A. de Meijere, *Org. Lett.*, 2006, **8**, 3635–3638.
19 S. Grimme, M. Steinmetz and M. Korth, *J. Chem. Theory Comput.*, 2007, **3**, 42–45.
20 Y. Zhao and D. G. Truhlar, *Org. Lett.*, 2006, **8**, 5753–5755.
21 P. Winget and T. Clark, *J. Comput. Chem.*, 2004, **25**, 725–733.
22 T. P. M. Goumans, A. W. Ehlers, K. Lammertsma, E.-U. Würthwein and S. Grimme, *Chem.–Eur. J.*, 2004, **10**, 6468–6475.
23 L. A. Curtiss, P. C. Redfern and K. Raghavachari, *J. Chem. Phys.*, 2005, **123**, 124107–12.
24 B. J. Lynch and D. G. Truhlar, *J. Phys. Chem. A*, 2003, **107**, 8996–8999.
25 Y. Zhao and D. G. Truhlar, *J. Phys. Chem. A*, 2006, **110**, 10478–10486.
26 Y. Zhao and D. G. Truhlar, *Theor. Chem. Acc.*, 2008, **120**, 215–241.
27 L. Goerigk and S. Grimme, *J. Chem. Theory Comput.*, 2010, **6**, 107–126.
28 S. Grimme, M. Steinmetz and M. Korth, *J. Org. Chem.*, 2007, **72**, 2118–2126.
29 M. P. Repasky, J. Chandrasekhar and W. L. Jorgensen, *J. Comput. Chem.*, 2002, **23**, 1601–1622.
30 K. W. Sattelmeyer, J. Tirado-Rives and W. L. Jorgensen, *J. Phys. Chem. A*, 2006, **110**, 13551–13559.
31 R. Ahlrichs, M. Bär, M. Häser and H. Horn, *Chem. Phys. Lett.*, 1989, **162**, 165–169.
32 R. Ahlrichs, *et al.*, TURBOMOLE 5.9, Universität Karlsruhe, 2008. See http://www.turbomole.com.
33 S. Grimme, RICC: A coupled-cluster program using the RI approximation, University of Münster, 2007.
34 A. Schäfer, C. Huber and R. Ahlrichs, *J. Chem. Phys.*, 1994, **100**, 5829–5835.
35 F. Weigend, F. Furche and R. Ahlrichs, *J. Chem. Phys.*, 2003, **119**, 12753–12762.
36 T. H. Dunning, Jr., *J. Chem. Phys.*, 1989, **90**, 1007–1023.
37 C. Møller and M. S. Plesset, *Phys. Rev.*, 1934, **46**, 618–622.
38 M. Pitoňák, P. Neogrády, J. Černý, S. Grimme and P. Hobza, *ChemPhysChem*, 2009, **10**, 282–289.
39 A. D. Becke, *Phys. Rev. A: At., Mol., Opt. Phys.*, 1988, **38**, 3098–3100.
40 C. Lee, W. Yang and R. G. Parr, *Phys. Rev. B: Condens. Matter*, 1988, **37**, 785–789.
41 J. P. Perdew, *Phys. Rev. B: Condens. Matter*, 1986, **33**, 8822–8824.

42 J. P. Perdew, K. Burke and M. Ernzerhof, *Phys. Rev. Lett.*, 1996, **77**, 3865–3868.
43 J. Tao, J. P. Perdew, V. N. Staroverov and G. E. Scuseria, *Phys. Rev. Lett.*, 2003, **91**, 146401–146401-4.
44 A. D. Becke, *J. Chem. Phys.*, 1993, **98**, 5648–5652.
45 P. J. Stephens, F. J. Devlin, C. F. Chabalowski and M. J. Frisch, *J. Phys. Chem.*, 1994, **98**, 11623–11627.
46 C. Adamo and V. Barone, *J. Chem. Phys.*, 1999, **110**, 6158–6170.
47 A. D. Boese and J. M. L. Martin, *J. Chem. Phys.*, 2004, **121**, 3405–3416.
48 Y. Zhao and D. G. Truhlar, *J. Phys. Chem. A*, 2005, **109**, 5656–5667.
49 T. Schwabe and S. Grimme, *Phys. Chem. Chem. Phys.*, 2006, **8**, 4398–4401.
50 A. Karton, A. Tarnopolsky, J.-F. Lamère, G. C. Schatz and J. M. L. Martin, *J. Phys. Chem. A*, 2008, **112**, 12868–12886.
51 T. Schwabe and S. Grimme, *Phys. Chem. Chem. Phys.*, 2007, **9**, 3397–3406.
52 K. Eichkorn, O. Treutler, H. Öhm, M. Häser and R. Ahlrichs, *Chem. Phys. Lett.*, 1995, **240**, 283–289.
53 K. Eichkorn, F. Weigend, O. Treutler and R. Ahlrichs, *Theor. Chem. Acc.*, 1997, **97**, 119–124.
54 F. Weigend, A. Köhn and C. Hättig, *J. Chem. Phys.*, 2002, **116**, 3175–3183.
55 P. Jurečka, J. Šponer, J. Černy and P. Hobza, *Phys. Chem. Chem. Phys.*, 2006, **8**, 1985–1993.
56 J. March, *Advanced Organic Chemistry*, Wiley Interscience Publication, New York, 4th edn, 1992.
57 F. A. Carey, R. J. Sundberg, H. J. Schäfer, D. Hoppe and G. Erker, *Organische Chemie. Ein weiterführendes Lehrbuch*, VCH, Weinheim, 1995.
58 H. Beyer, W. Walter and W. Francke, *Lehrbuch der Organischen Chemie*, S. Hirzel Verlag, Stuttgart, 23 edn, 1998.
59 J. Clayden, N. Greeves, S. Warren and P. Wothers, *Organic Chemistry*, Oxford University Press, Oxford, 2001.
60 J. McMurry, *Organic Chemistry*, Brooks/Cole Publication Company, 4th edn, 1995.
61 F. Vögtle, *Reizvolle Moleküle in der Organischen Chemie*, Teubner Verlag, 1989.
62 R. Brückner, *Reaktionsmechanismen*, Spektrum Akademischer Verlag, München, 3 edn, 2004.
63 S. M. Kupchan, C.-K. Kim and K. Miyano, *J. Chem. Soc., Chem. Commun.*, 1976, 91–92.
64 S. Kim, B. Liu, C. Park and K. Lee, *J. Org. Chem.*, 1998, **63**, 1571–1573.
65 R. Orugunty, I. Ghiviriga, K. Abboud, M. Battiste and D. Wright, *J. Org. Chem.*, 2004, **69**, 570–572.
66 C. Ye and J. M. Shreeve, *J. Fluorine Chem.*, 2004, **125**, 1869–1872.
67 K. Crawford, S. Bur, C. Straub and A. Padwa, *Org. Lett.*, 2003, **5**, 3337–3340.
68 J. A. Barten, A. A. Kadyrov and G.-V. Roschenthaler, *J. Fluorine Chem.*, 2002, **113**, 101–103.
69 S. Starling, S. Vonwiller and J. Reek, *J. Org. Chem.*, 1998, **63**, 2262–2272.
70 G. Blay, A. M. Collado, B. Garcia and J. R. Pedro, *Tetrahedron*, 2005, **61**, 10853–10860.
71 http://www.uni-muenster.de/Chemie.oc/grimme/en/index.html see GMTKN24 database.
72 T. Helgaker, W. Klopper, H. Koch and J. Noga, *J. Chem. Phys.*, 1997, **106**, 9639–9646.
73 P. Pyykkö, *Chem. Rev.*, 1997, **97**, 597–636.
74 S. M. Cybulski and M. L. Lytle, *J. Chem. Phys.*, 2007, **127**, 141102.
75 S. Grimme, *Chem.–Eur. J.*, 2004, **10**, 3423–3429.
76 J.-D. Chai and M. Head-Gordon, *J. Chem. Phys.*, 2009, **131**, 174105.

 This journal is © the Owner Societies 2010